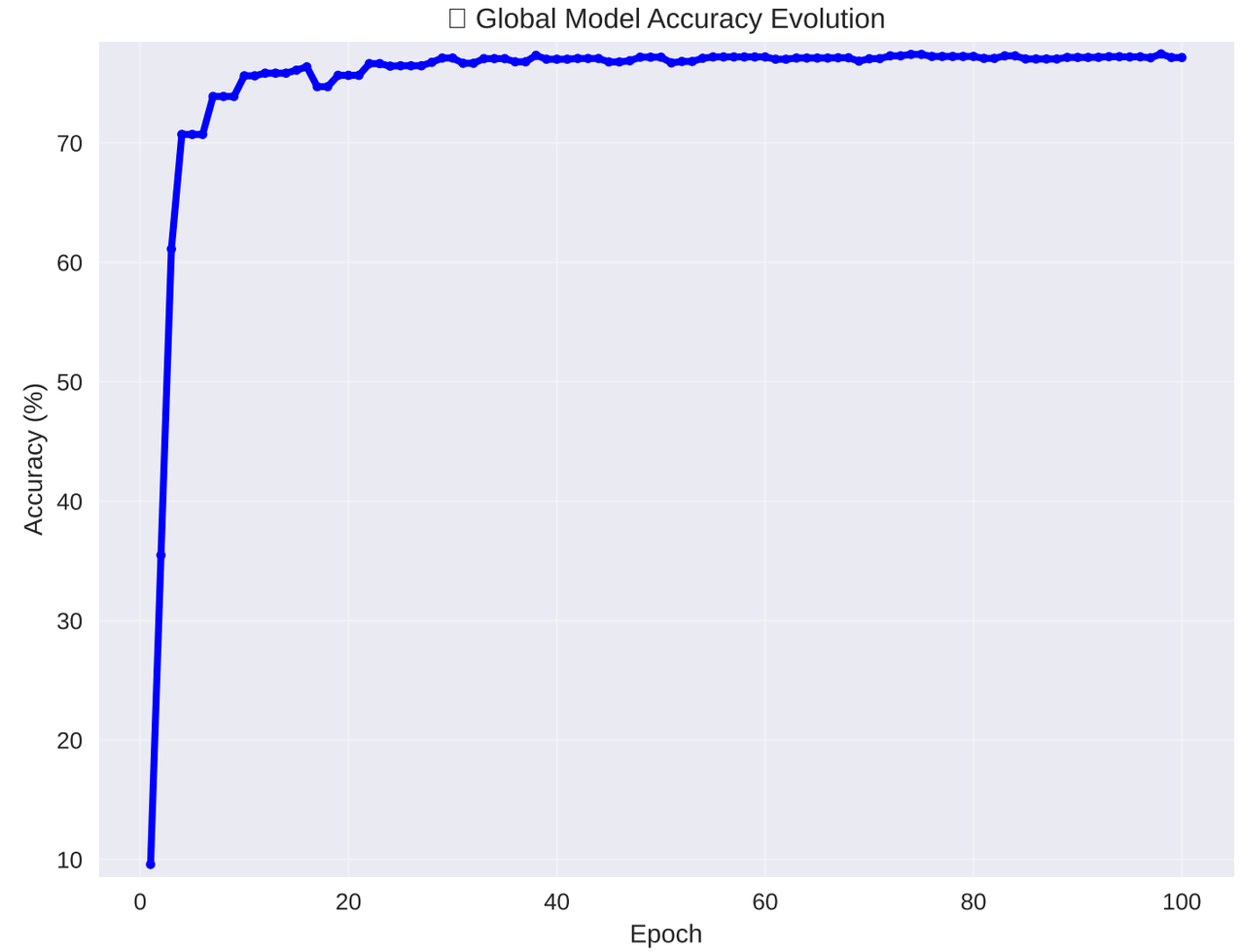
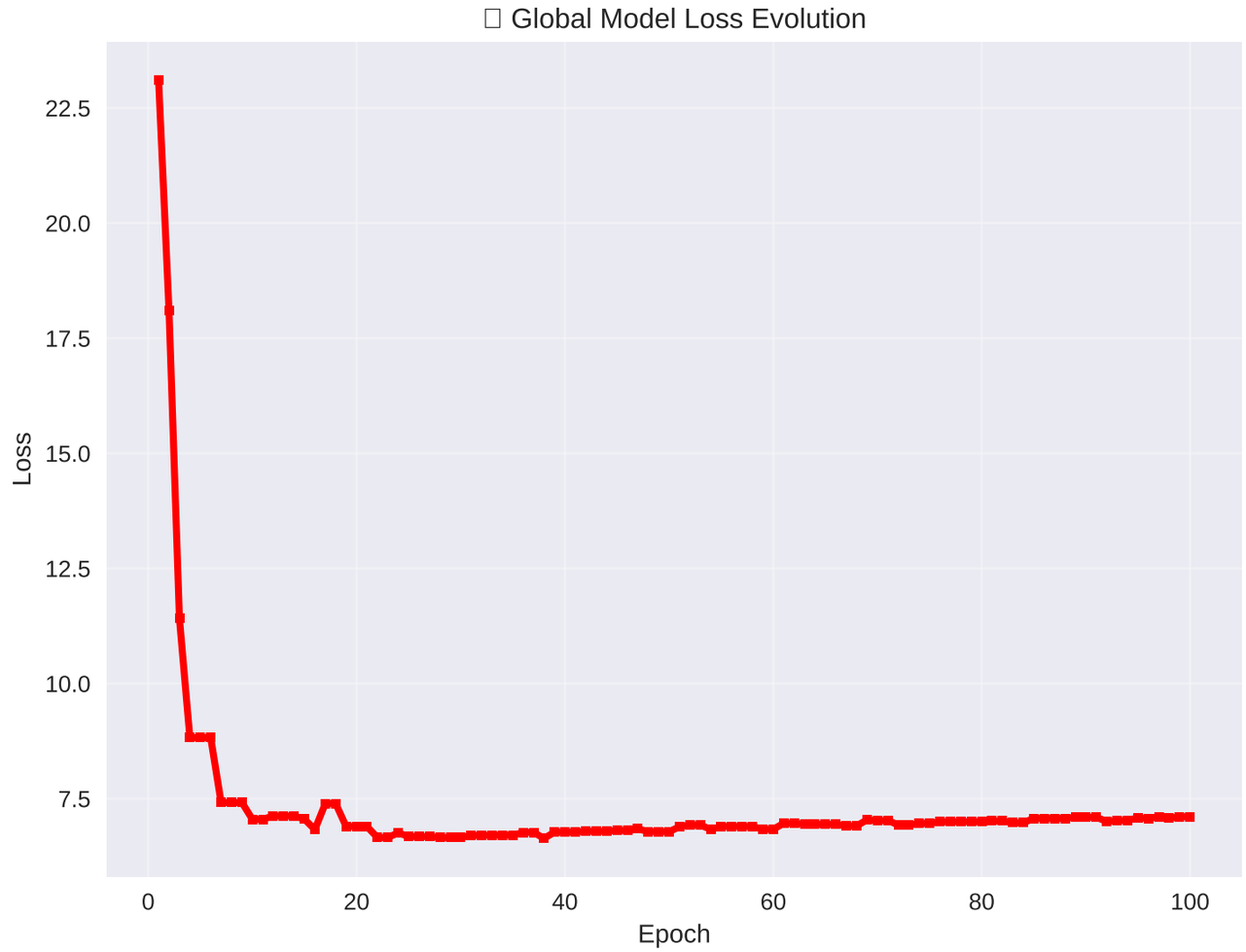
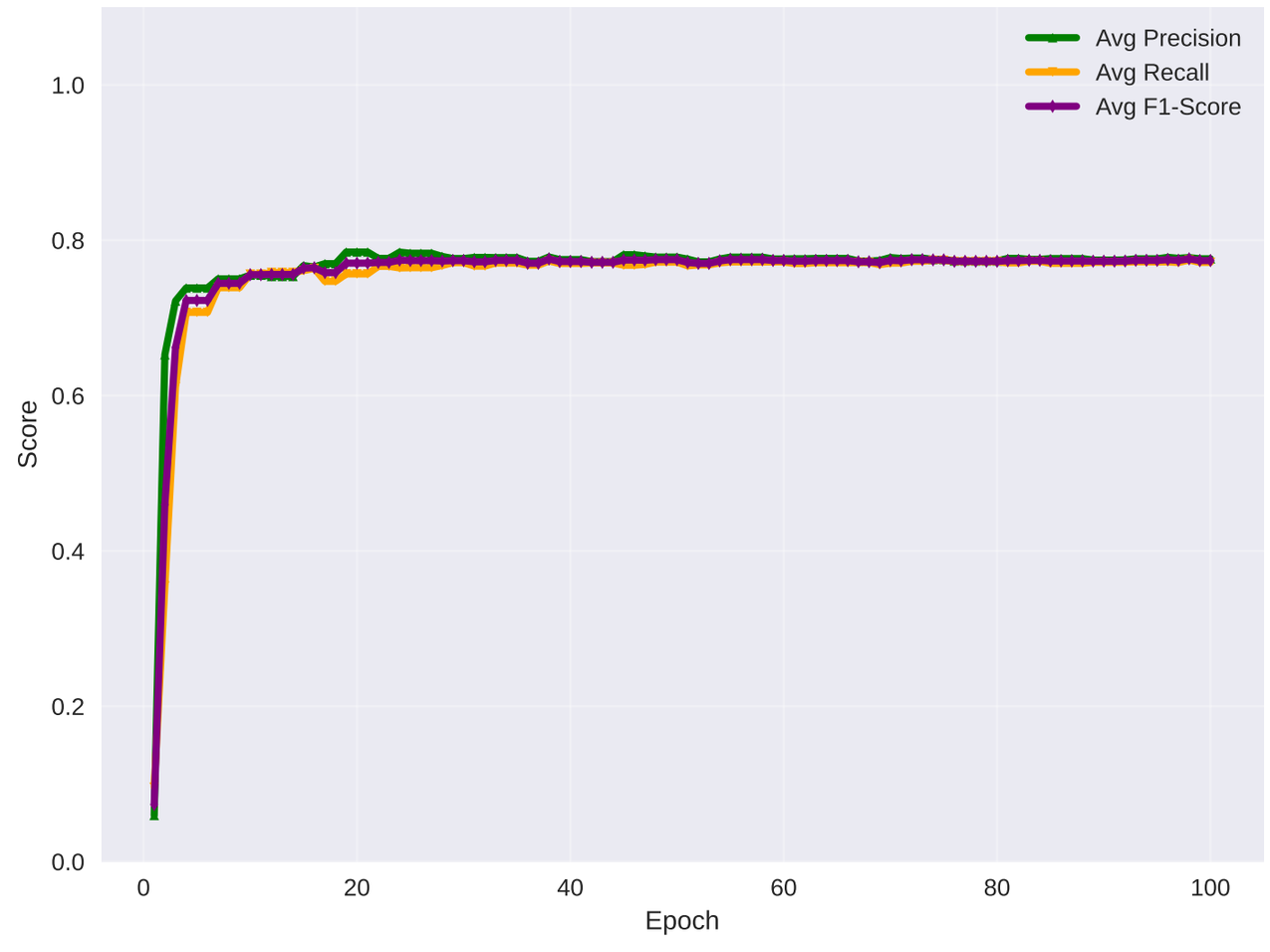
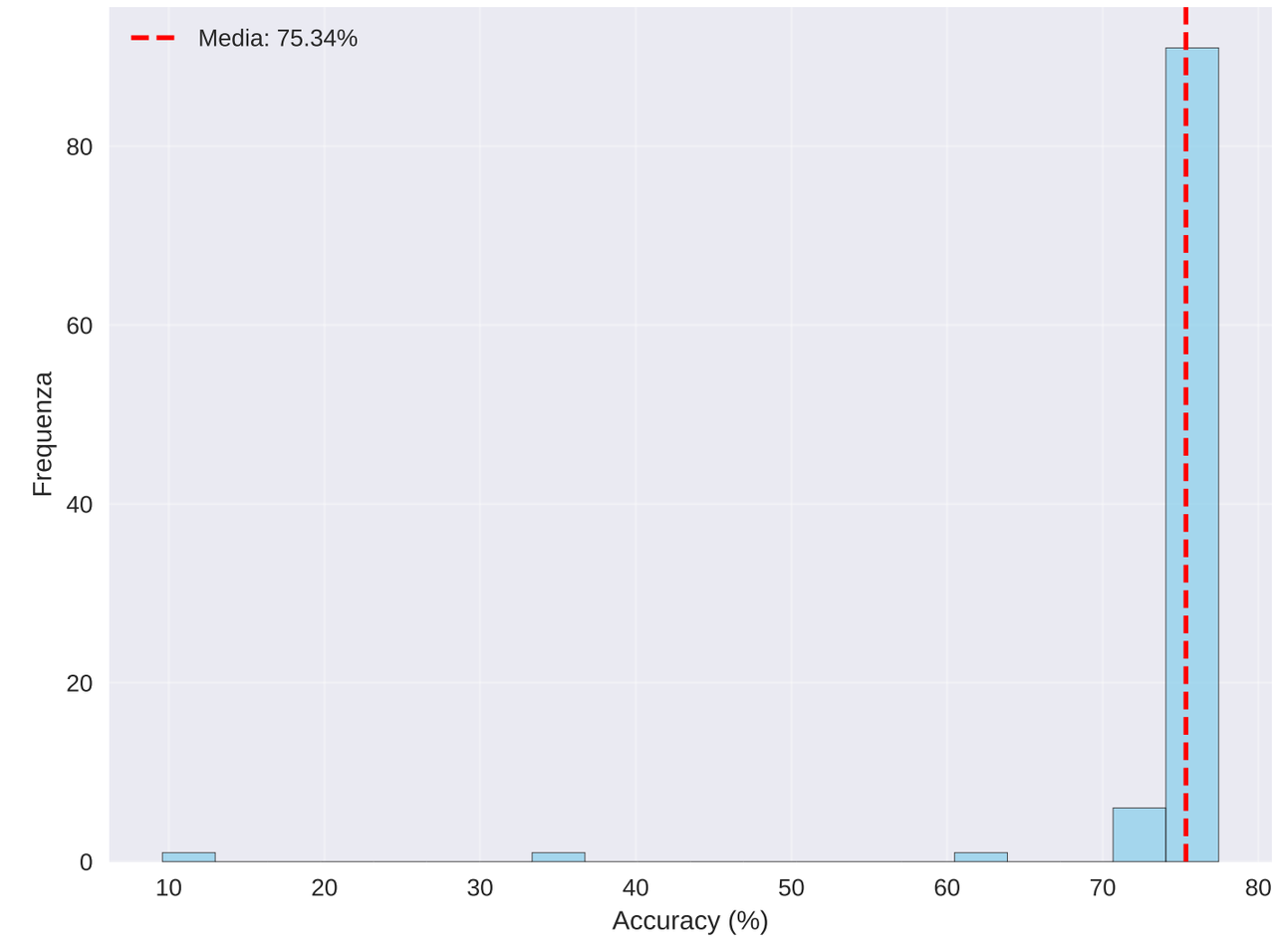
Federated Learning - Main Metrics Overview

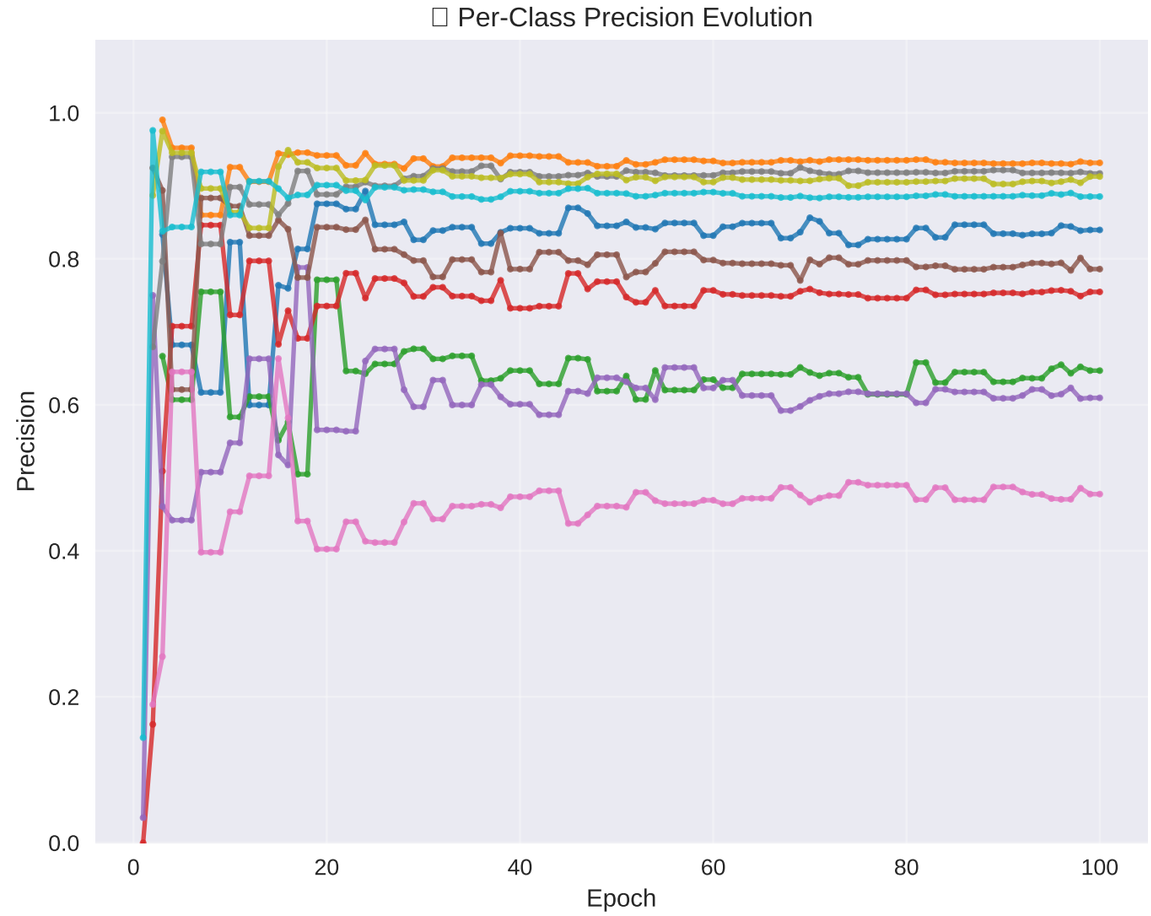


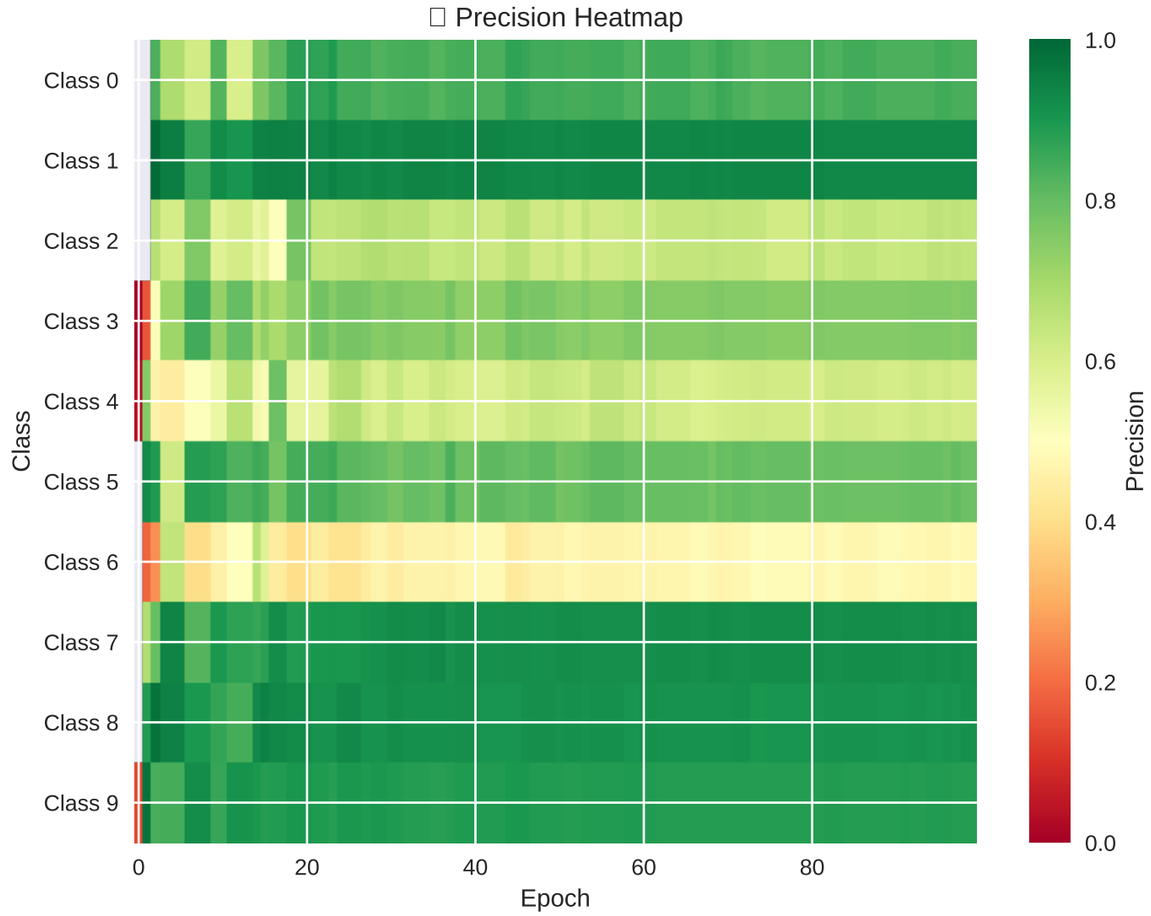


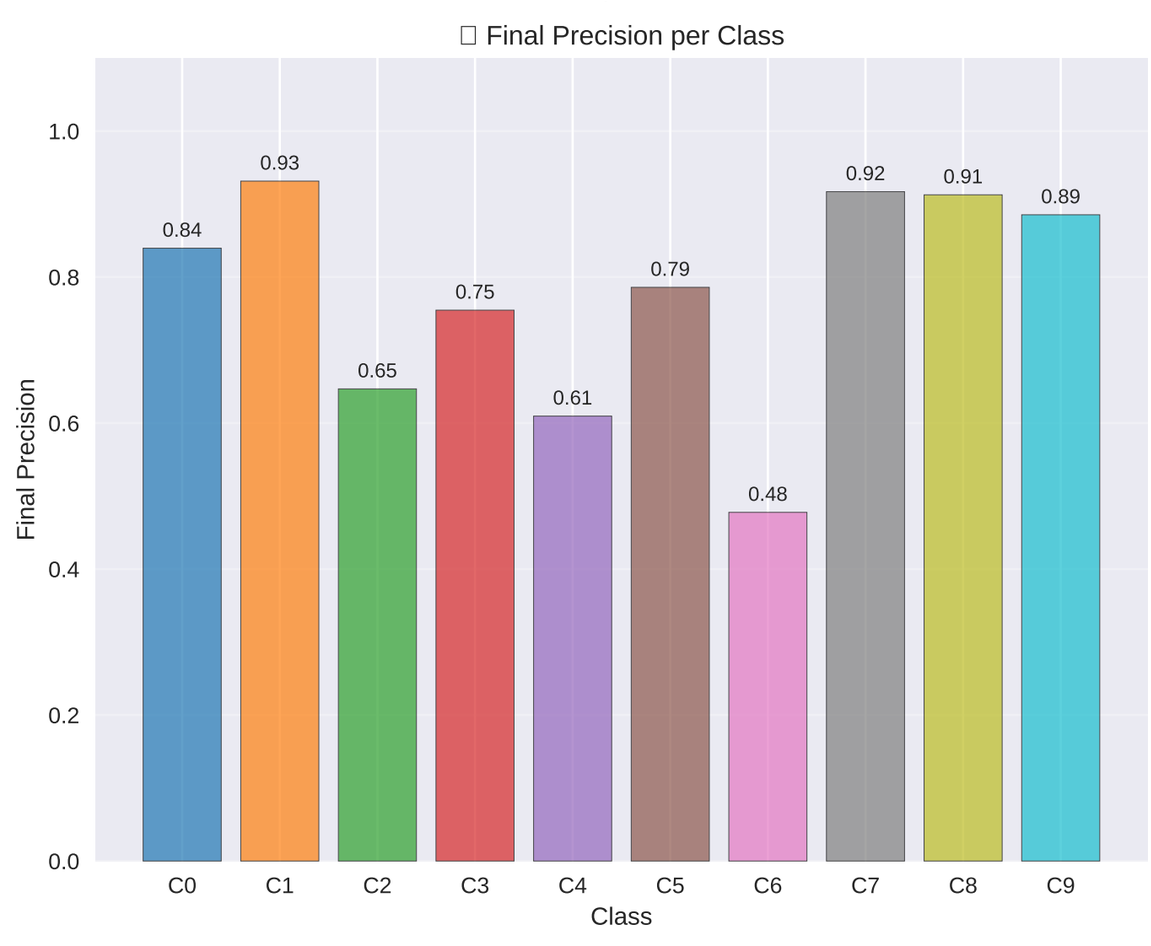


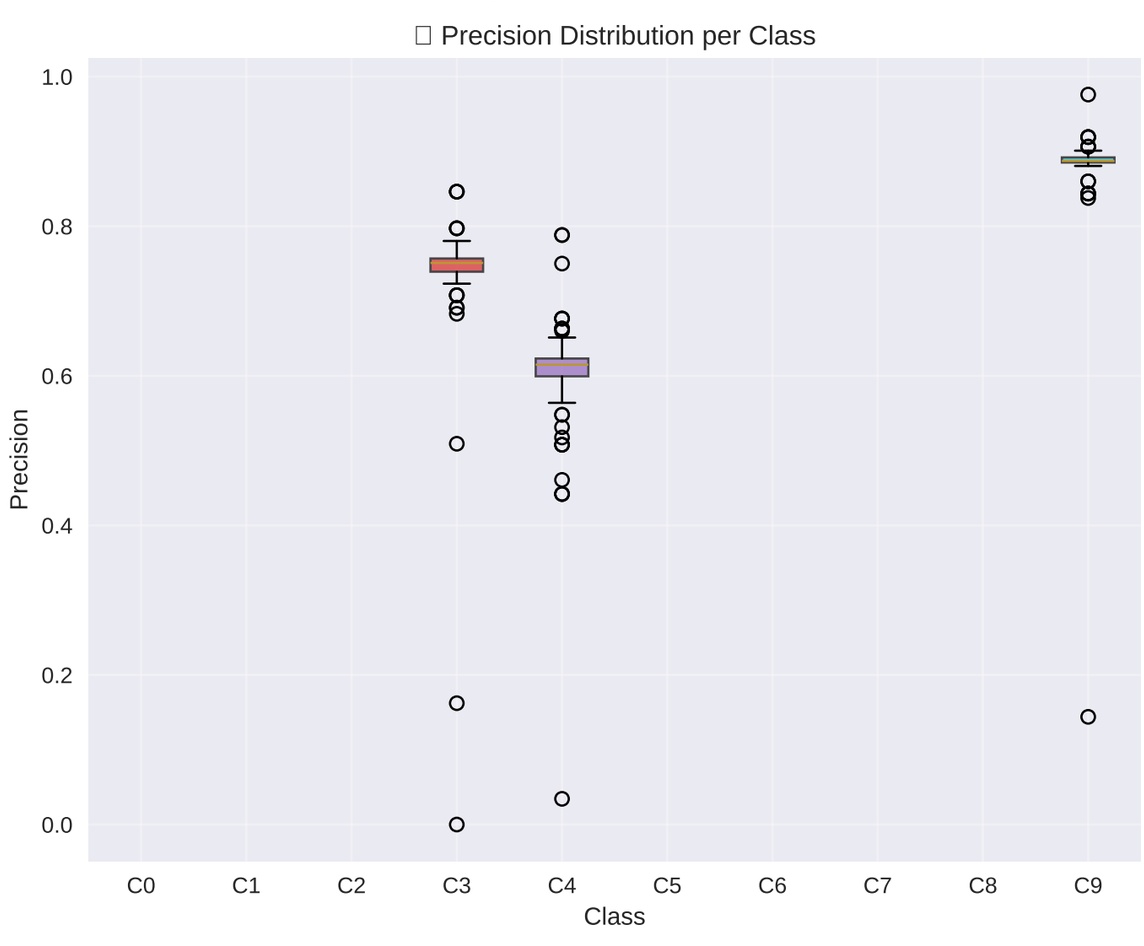


Per-Class Precision Analysis



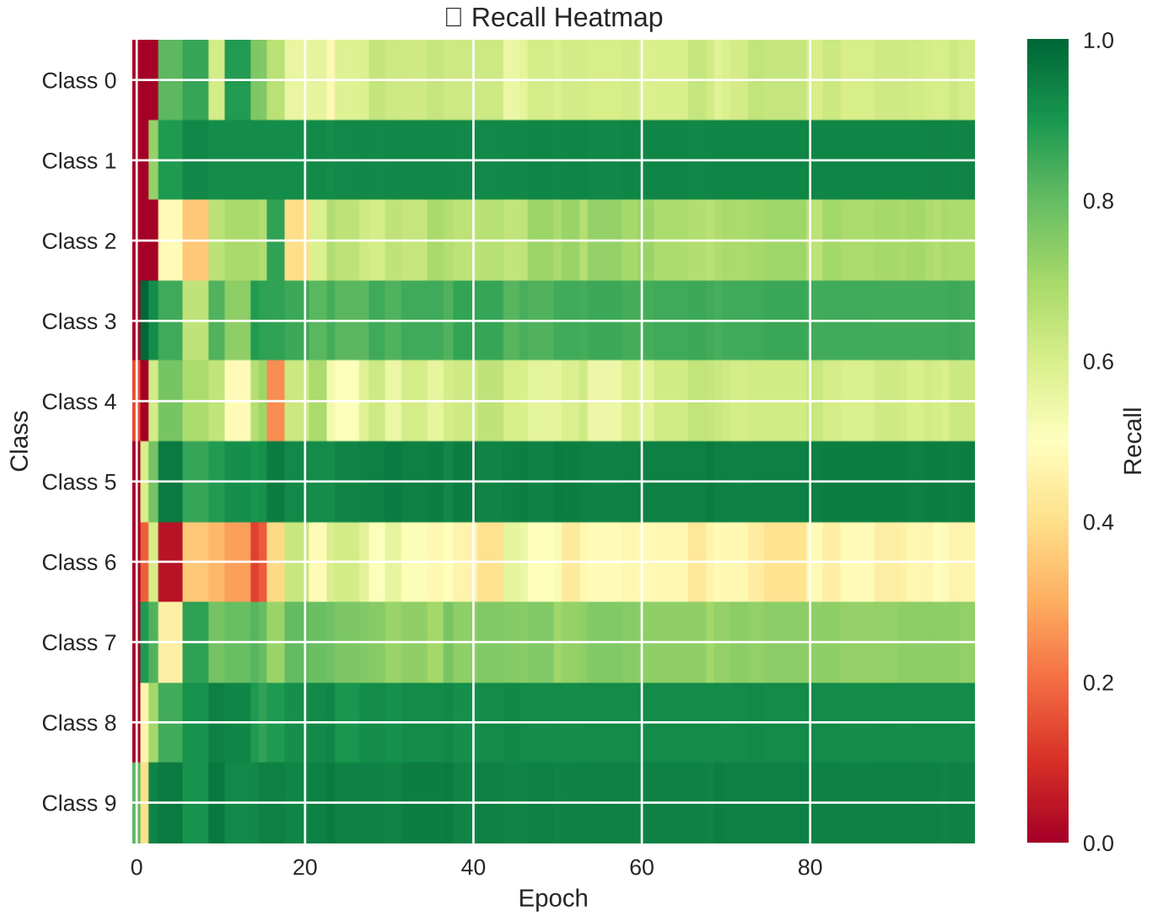


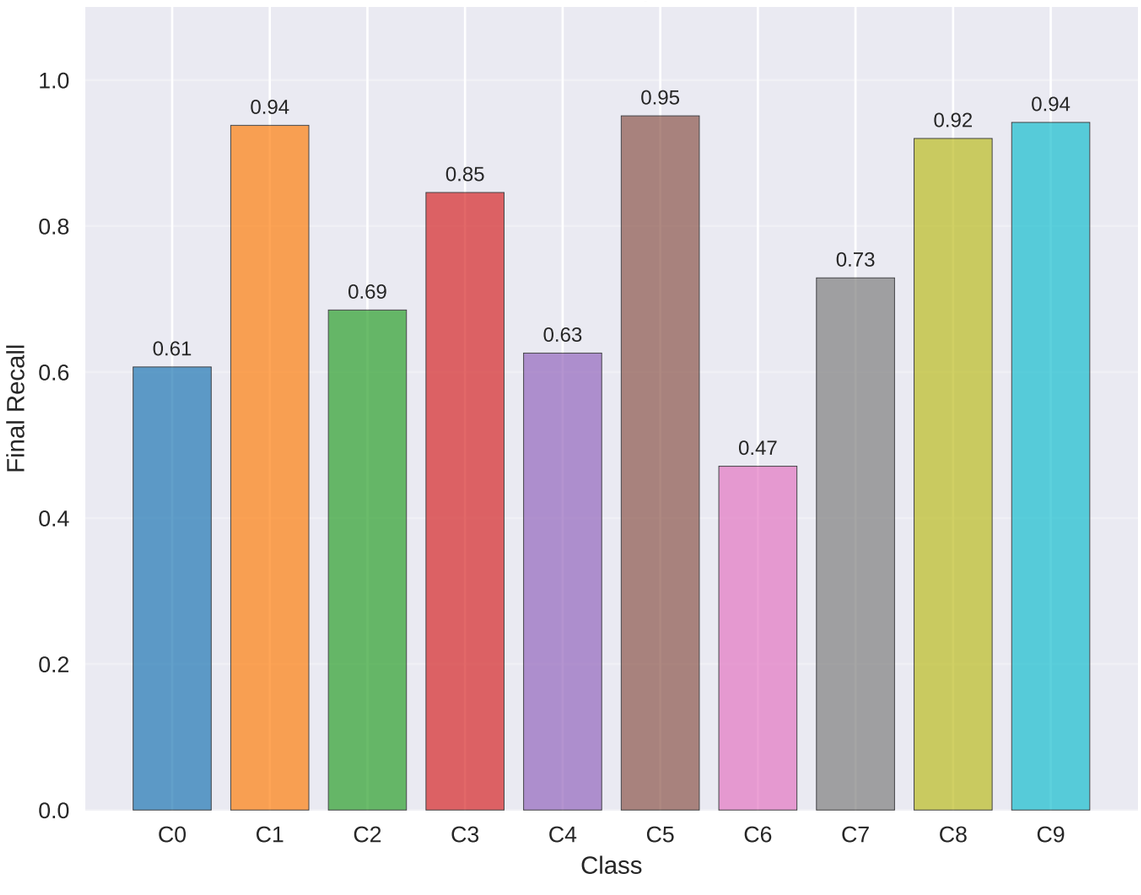


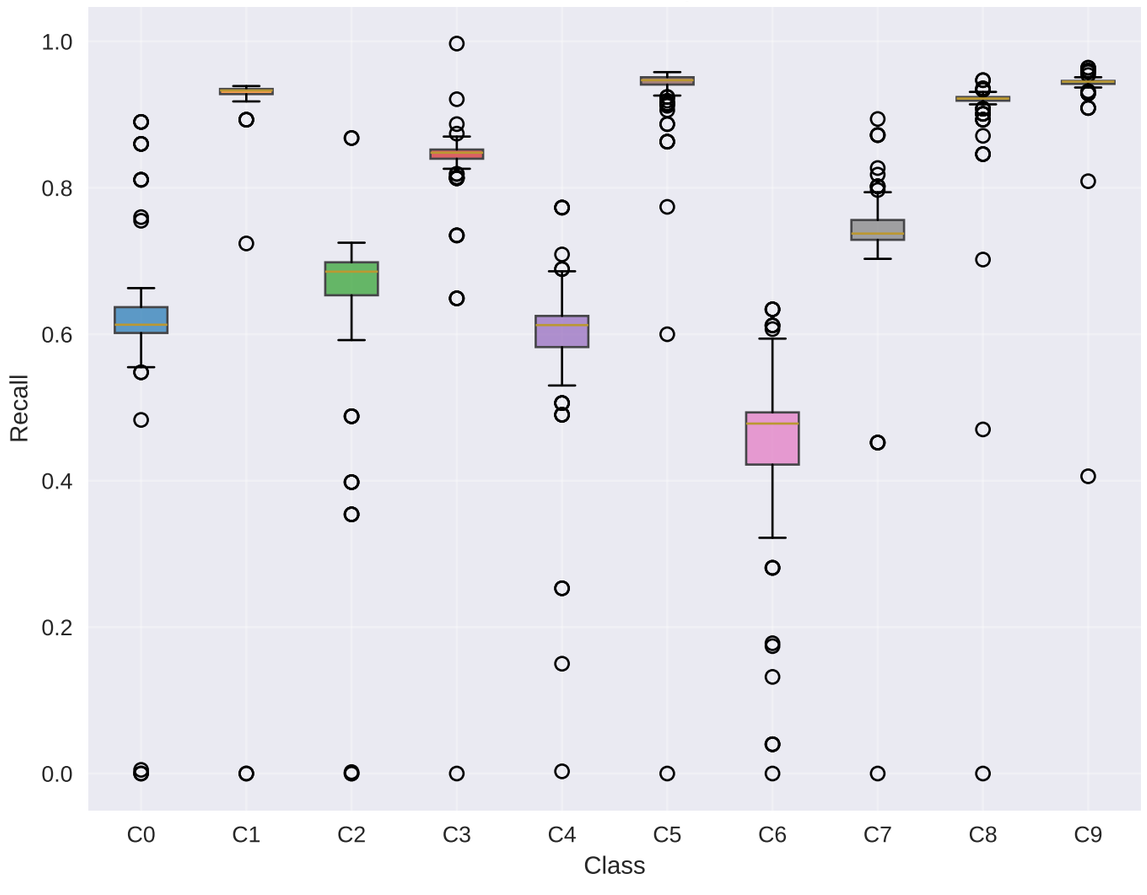


Per-Class Recall Analysis



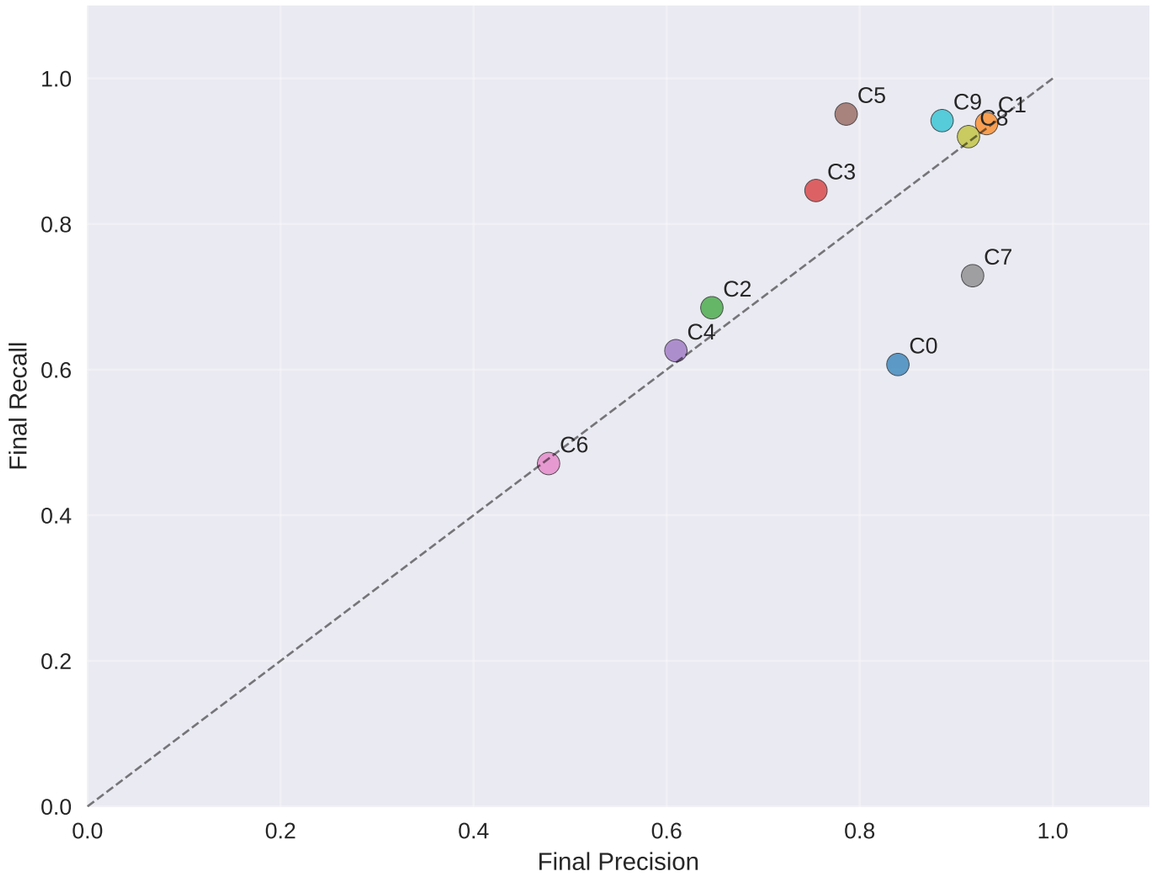




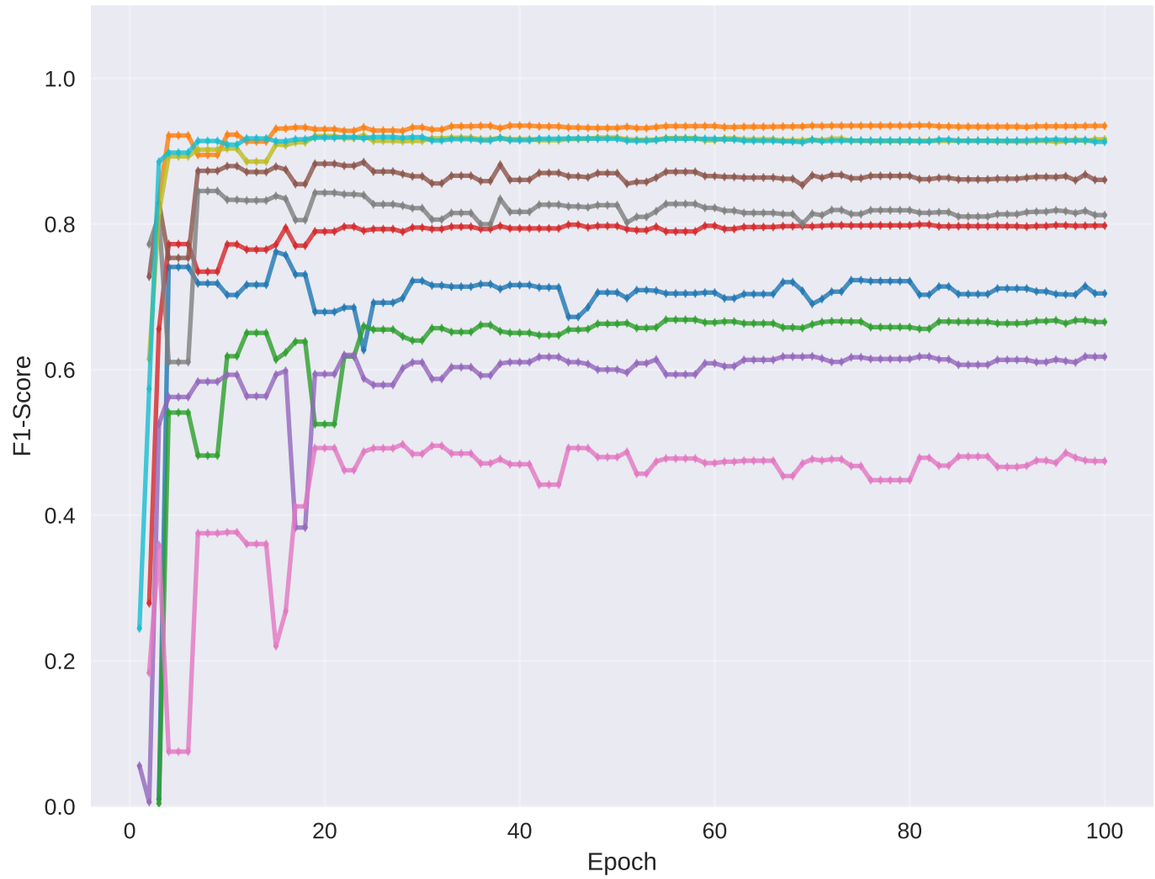


Comparative Analysis

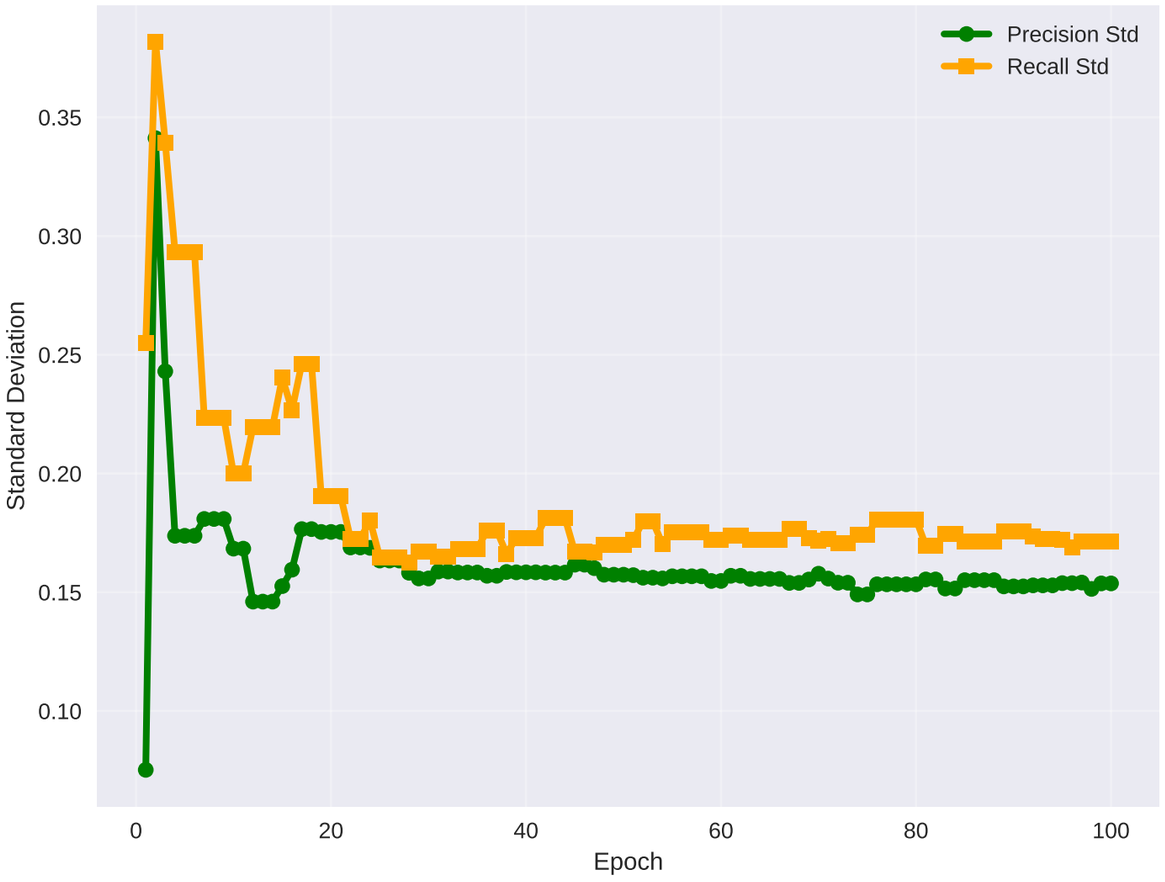
□ Precision vs Recall (Final Values)



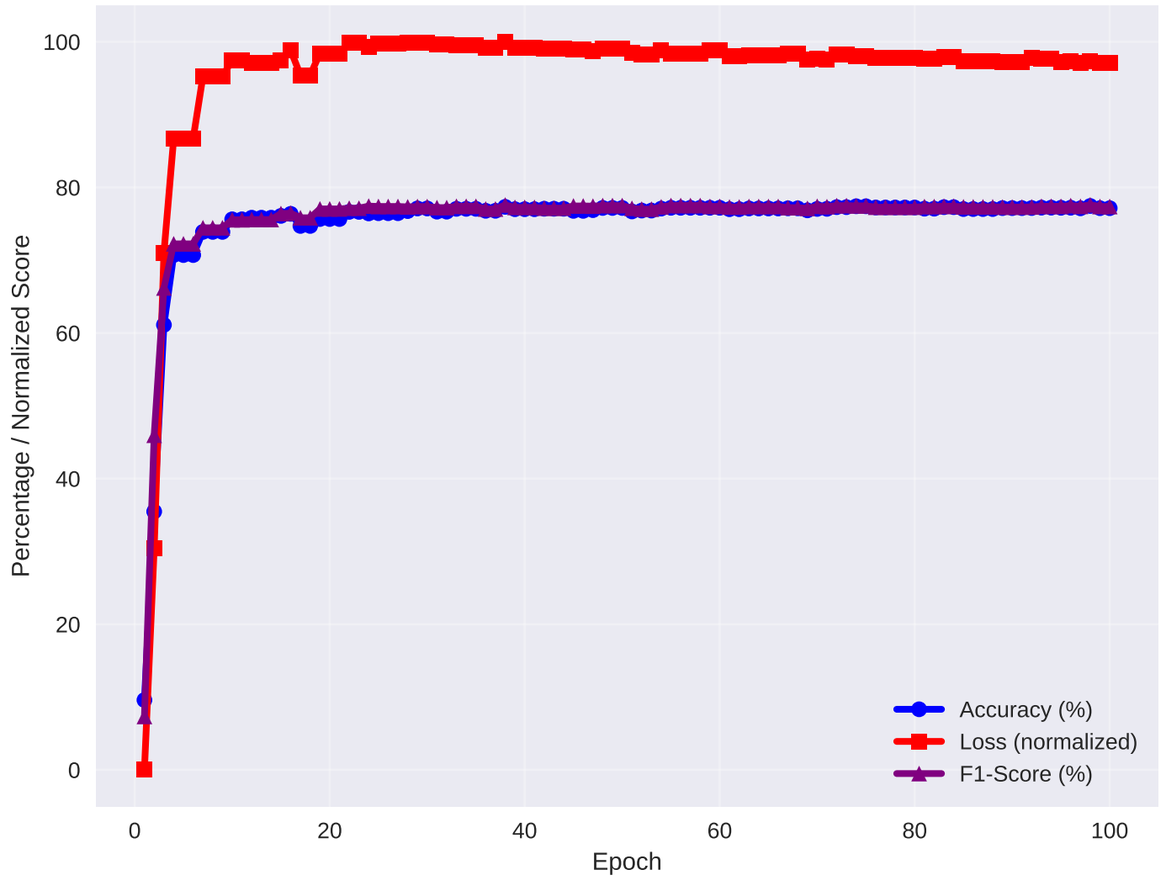
□ Per-Class F1-Score Evolution



□ Inter-Class Variance Evolution



□ Learning Curves Comparison



Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-09-24 15:51:55
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 77.15%
• Loss: 7.1062
• Average Precision: 0.776
• Average Recall: 0.771
• Average F1-Score: 0.774

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +67.56%
• Riduzione Loss: +16.0043
• Stabilità Accuracy (CV): 0.1069

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.931)
• Peggiore Precision: Classe 6 (0.478)
• Migliore Recall: Classe 5 (0.951)
• Peggiore Recall: Classe 6 (0.471)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.840, R=0.607, F1=0.705
• Classe 1: P=0.931, R=0.938, F1=0.935
• Classe 2: P=0.647, R=0.685, F1=0.665
• Classe 3: P=0.755, R=0.846, F1=0.798
• Classe 4: P=0.610, R=0.626, F1=0.618
• Classe 5: P=0.786, R=0.951, F1=0.861
• Classe 6: P=0.478, R=0.471, F1=0.474
• Classe 7: P=0.917, R=0.729, F1=0.812
• Classe 8: P=0.913, R=0.920, F1=0.916
• Classe 9: P=0.885, R=0.942, F1=0.913

VALUTAZIONE QUALITATIVA:
• Performance Globale: DA MIGLIORARE
• Stabilità Training: INSTABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
• Considerare aumento epoche o tuning iperparametri
• Implementare class balancing o weighted loss
• Migliorare stabilità con learning rate scheduling
```